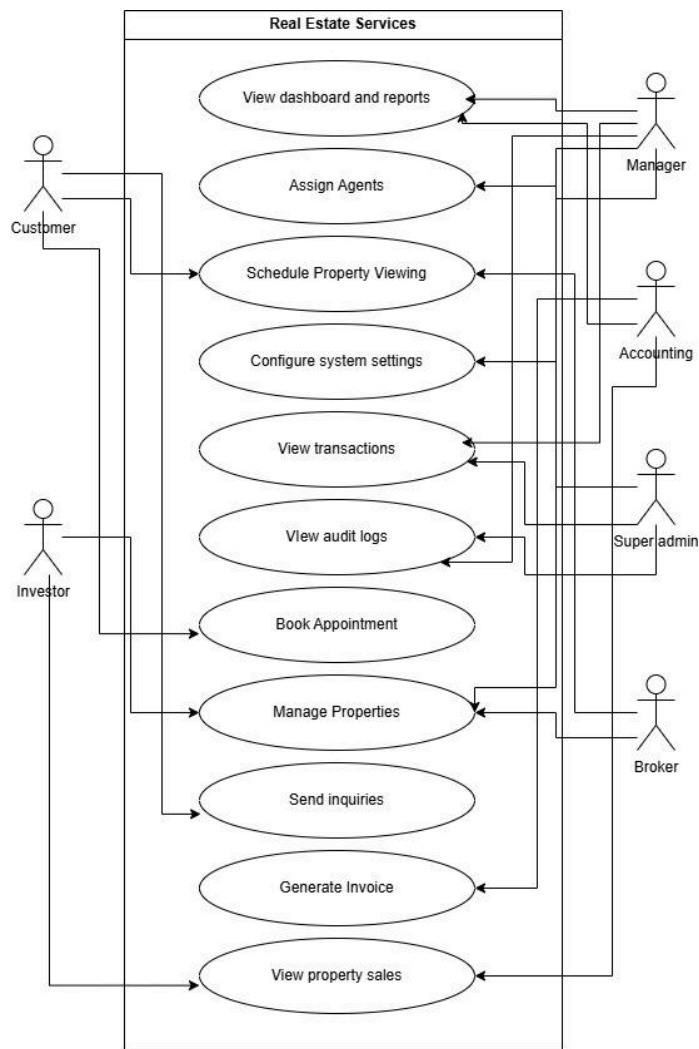


NAME	Stephen Adrien S. Tatel
PROJECT TITLE	EstateFlow: A Web-Based Customer Relationship and Financial Management System for Real Estate
SUBJECT: CODE: TIME:	IT15/L Integrative Programming and Technologies
TOPIC (Type of Business Process)	#5 Customer Relationship and Engagement System(CRM+)
Products/Services	Real Estate Services
Tools Needed:	Visual Studio 2022, SSMS,
Technology:	PayMongo API, SMTP, ReCaptcha,
Security Features	Authentication and Authorization (Role-Based Access Control) Input Validation (Prevent SQL Injection & XSS) Password Hashing (bcrypt) Audit Logging and Monitoring Session Management
Target user/s:	Super admin,Company Manager, Agent/Broker, Investor, Customer, Accounting Staff
Subs System / Management Transaction/Modules	1.Customer Master Data Management 2.Sales Pipeline and Opportunity Tracking 3.Customer Support and Ticketing 4. Marketing Campaign and Automation 5.Billing and Customer Financial Integration 6. Payroll and Commission Integration 7. Geospatial Master Data Management 8. Digital Document and E-Signature Vault 9. Smart Routing and Site-Trip Management

LOGO	
Project Objectives:	• To develop a web-based system that centralizes customer, sales, and financial data for real estate transactions within one platform. • To implement a sales pipeline tracking system that improves the monitoring of customer interactions and deal progress. • To integrate billing and financial management for accurate tracking of transactions, payments, and commissions.
Project Description:	This project aims to develop a web-based real estate management system that integrates Customer Relationship Management (CRM) and Financial Management into a centralized platform. It helps real estate businesses manage customer interactions, property transactions, billing, and commission tracking efficiently. The system is designed for administrators, agents, customers, and property investors. Admins manage operations and approvals, agents handle sales and client interactions, customers browse and transact properties, and investors list properties and monitor earnings. It addresses issues such as disorganized data, poor transaction tracking, and inefficient communication. By centralizing these processes, the system improves efficiency, enhances user experience, and supports better decision-making. The system will be developed using web technologies such as ASP.NET, SQL Server, ensuring scalability, reliability, and security through features like authentication and audit logging.

Use Case Diagram (Role-Based Access)



Role-Based Access for Real Estate Services

Super admin

- View audit logs
- View transactions
- Configure system

Company Manager

- Monitor sales pipeline
- Assign agents to properties
- View transactions and reports
- Oversee daily operations

Agent/Broker

- Manage customers
- Track sales pipeline
- Schedule property viewings

	● Handle inquiries and close deals Customer/Client ● Browse and search properties ● Send inquiries ● Book appointments ● View transaction status Investor ● Add and manage properties ● Choose selling or renting options ● Track earnings and transactions Accounting ● Manage billing and payments ● Record financial transactions ● Generate financial reports
ERD(Entity Relationship Diagram)	<pre>graph TD Payment --> Invoice Invoice --> Transaction Transaction --> Commission Commission --> Agent Agent --> Client Client --> Appointment Appointment --> Leads Leads --> Properties Properties --> Investor Investor --> Appointment Appointment --> Users Users --> Roles Roles --> Users Users --> AuditLogs AuditLogs --> Users</pre> The ERD illustrates the following entities and their attributes: Payment: PaymentID (PK), InvoiceID, Amount, PaymentDate, PaymentMethod, ReferenceNumber Invoice: InvoiceID (PK), TransactionID, Amount, DueDate, Status Properties: PropertyID (PK), InvestorID (FK), Title, Description, Address, Listing_mode, Price, Status Investor: InvestorID (PK), Fname, Mname, Lname Leads: LeadID (PK), ClientID (FK), PropertyID (FK), AgentID (FK), Status, AgentID (FK) Appointment: AppointmentID (PK), AgentID (FK), LeadID, DateTime, Status, Note, Note Users: UserID (PK), Fullname, Email, Password, Confirm Password, Role Roles: RolesID (PK), RoleType Agent: AgentID (PK), LicenseNumber, Specialization, CommissionRate, Status Client: ClientID (PK), Fname, Mname, Lname Transaction: TransactionID (PK), Amount, DueDate, Status Commission: CommissionID (PK), TransactionID (FK), AgentID (FK), Amount, Status Audit Logs: LogsID (PK), UserID (FK), Action, Timestamp, IP Address
Date Submitted:	March 30, 2026
Teacher's Feedback	

	Student's Signature (after feedbacking) Teacher's Signature

